

Task - 1.1 & 1.2: Neural Network Implementation and Pre-Processing Analysis

3-Layer FCNN on UCI Adult Census Income

1 Introduction

This report documents the end-to-end implementation of a 3-layer Fully Connected Neural Network (FCNN) built from scratch using NumPy. The study is divided into two phases: the initial architectural setup and a comparative analysis of how input data scaling affects gradient flow and model convergence.

2 Baseline Configuration

The following parameters were established for the core engine:

- **Architecture:** 3-layer FCNN ($108 \rightarrow 64 \rightarrow 32 \rightarrow 1$).
- **Activations:** ReLU for hidden layers, Sigmoid for output.
- **Initialization:** He initialization ($w = \text{np.random.randn} \cdot \sqrt{2/n}$).
- **Optimizer:** Vanilla SGD (Learning Rate: 0.01).
- **Loss Function:** Binary Cross-Entropy.
- **Iterations:** 1,000.

3 Implementation & Experimentation

The project focused on manual backpropagation and the critical role of data preparation.

3.1 From-Scratch Implementation (Task 1.1)

- **Forward Pass:** Implemented using dot products and manual activation functions.
- **Backpropagation:** Gradients derived via the chain rule (e.g., $dz_3 = a_3 - Y$).
- **Results:** The model achieved steady improvement, reaching a final accuracy of **76.39%**, exceeding the 75% requirement.

3.2 Pre-Processing Experiments (Task 1.2)

To test the impact of scaling, the model was trained on two different data states:

Test A: Raw Data

- **Expectation:** Large feature scales (e.g., "Capital Gain") will cause gradient imbalance.
- **Results:** Extremely high initial loss (13.9848) and slow progress after an initial drop.
- **Observation:** The model reached a suboptimal accuracy of 75.92%.

Test B: Min-Max Scaled Data

- **Change:** All features scaled to $[0, 1]$.
- **Results:** Much lower initial loss (0.8725) and consistent convergence.
- **Observation:** Final accuracy improved to **82.21%**.

4 Comparative Performance

Table 1: Impact of Scaling on Training

Configuration	Initial Loss	Final Loss	Final Accuracy
Raw Data	13.9848	0.5551	75.92%
Min-Max Scaled	0.8725	0.3967	82.21%

5 Impact Analysis

1. **Gradient Flow:** In raw data, features with massive ranges dominate updates, while binary features are ignored.
2. **Convergence Speed:** Scaled data converged **~9× faster** than raw data.
3. **Optimization Landscape:** Scaling transforms the loss surface from an elongated elliptical shape to a more spherical one, allowing SGD to progress efficiently.

6 Conclusion

The experiments demonstrate that while the FCNN engine is mathematically sound, its performance is strictly tied to data quality.

- **Accuracy Gain:** Pre-processing provided a +6.29% absolute improvement.
- **Stability:** Scaling prevents the "plateau effect" and results in a 28.5% reduction in final loss.

- **Recommendation:** Feature scaling is non-negotiable for neural network training.